TECHNICAL SUPPLEMENT 09 · ENGLISH

Data Provenance and Numerical Reproducibility

Agri-Social Navigation in Vineyard Corridors

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Study identity	Specification
Campaign / Campaña	EVAL_20261019_112257
Navigation / Navegación	9 × 4 × 10 = 360; SUCCESS 266; TIMEOUT 94
Benchmark	180; 20 per method / por método
Version / Versión	20 October / octubre 2026

Numerical companion to the revised Sensors manuscript. This report uses the identified endpoint dataset and separates it from the older public campaigns. All seven authors are listed in the original manuscript order.

1. Reading documents versus numerical inputs

The documentation index links English and Spanish PDF reports with all seven coauthors. CSV/JSON are retained under data/; results/expected/ for machine-readable input and audit; they are not Excel reports. No Overleaf ZIP is required in the public repository.

2. Reproduction commands

```
python -m pip install -r reproducibility/requirements.txt
python -B reproducibility/reproduce.py --out reproduced
python -B reproducibility/verify.py --results reproduced
python -B reproducibility/verify_manifest.py
```

The independent checker verifies accounting, contrasts, intervals/effects, Holm, decision matrices, equal-weight TOPSIS and dominance. reproduce.py also refits the reported models and regenerates resampling and sensitivity analyses. Neither program executes Unity or recalculates raw navigation metrics. The included analysis regenerates the numerical results from per-run metrics.

3. Campaign separation and preservation

Only the identified current campaign EVAL_20261019_112257 is packaged. Historical campaigns, backups and legacy folders are excluded. Its 360 attempts comprise 266 SUCCESS and 94 TIMEOUT outcomes; a TIMEOUT is terminal, not pending.

4. Executable scope and provenance

The release includes selected evaluated C# snapshots, controller configurations, human initial conditions and numerical inputs. Raw telemetry and the full Unity project are excluded. The package includes per-run metrics, numerical analysis, controller sources and configurations. The source freeze identifier is 104a6ca991e33019cd403aa019621f0cb8071d9f05906364d99e547efd499616. SHA256 manifests identify the supplied files; identity checks do not establish scientific validity by themselves.

All report authors follow the manuscript order. Vasconez.pdf and six original reference PDFs were recovered unchanged from the preceding repository revision. No coauthor approval, external peer-review acceptance or new experimental result is asserted by this documentation update.

Source and scope

Source: data/; results/expected/. PDFs provide the reading reports; CSV/JSON retain numerical inputs and provenance. Scope: evaluated simulation, without new experiments, independent raw-stream reconstruction or physical validation.